

MIDHUN PRAHASH SR

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Professional Summary

AI/DS Learner with deep expertise in Large Language Models (LLMs), scalable RAG architectures, and model fine-tuning. Skilled at solving complex data retrieval challenges by architecting multi-tenant knowledge graphs and optimizing LLM orchestration. Proven ability to build and deploy low-latency microservices, transforming intricate data environments into intuitive, real-time conversational AI and context-aware agents.

Technical Skills

Languages & Infrastructure: Python, SQL, FastAPI, PostgreSQL, Supabase, WebSockets, Docker
AI & Machine Learning: Supervised Learning, Feature Engineering, Statistical Modeling, Scikit-learn, Computer Vision, Transformers, BERT, RNN's, LSTM, CNN's
NLP & LLM Systems: RAG, GraphRAG, Semantic/Hybrid Search, Prompt Engineering, Few shot learning, Embedding Models, Tokenization Strategies, ASR/TTS
Data & Vector Stores: pgvector, FAISS, Neo4j, Knowledge Graphs, Pandas, NumPy, NLTK

Experience

MINDS Research Lab - IIIT Kottayam Jun 2026 – Present
Research Intern

- Researching advanced encoder-based transformer architectures to detect hidden suicidal intent within unstructured social media text, integrating sequential modeling and psycholinguistic feature fusion.

Cospin Jan 2026 – Mar 2026
AI Engineering Intern

- Engineered low-latency Python/FastAPI microservices to orchestrate real-time speech ingestion, LLM processing, and audio delivery within a conversational AI platform.
- Architected responsive ASR and TTS pipelines via WebSockets and Docker, optimizing inference workflows to enable seamless, human-like voice interactions.

Lightcast Sep 2025 – Jan 2026
AI/ML Intern

- Designed a GraphRAG hybrid search engine leveraging Neo4j to extract professional skills from unstructured job descriptions and map them to standardized taxonomy hierarchies.
- Implemented a 4-tier reranking pipeline (exact match, alias resolution, semantic, and contextual search) utilizing Gemini 2.5 Flash, drastically reducing hallucinations and boosting precision in ambiguous data mapping.

Orion Governance Jun 2025 – Jul 2025
Machine Learning Intern

- Deployed a supervised classification model utilizing fuzzy string matching and advanced feature engineering to detect unauthorized employee usernames with 89% accuracy.

Projects

UniChat | FastAPI, pgvector, LlamaIndex, Context-aware answering engine *Production RAG System*

- Architected a multi-tenant RAG backend integrating semantic caching, prompt orchestration, and Row-Level Security to stream citation-grounded answers across institutional datasets via Server-Sent Events (SSE).

StudentAI | Python, FAISS, Sentence Transformers, Gemini API *RAG Learning Assistant*

- Built an OCR-to-Vector pipeline to dynamically synthesize unstructured educational texts into targeted, evidence-backed conceptual explanations and learning assessments.

Review Analyzer | Selenium, NLTK, VADER, TextBlob *Sentiment Analysis Pipeline*

- Automated the extraction and NLP-driven sentiment evaluation of e-commerce feedback to uncover and visualize actionable user experience trends.

Hairline AI | YOLOv8, MediaPipe, OpenCV, SQL *Computer Vision Application*

- Engineered an automated facial analysis tool combining YOLOv8 segmentation and MediaPipe landmark detection to precisely quantify and store facial metric proportions.

Publications & Patents

Publication: Midhun Prahash SR, et al. "Enhancing Transformer-Based Hidden Suicidal Intention Detection with Sequential Modeling and Psycholinguistic Feature Fusion." IEEE ICITIT - 2026.

Patent: Application No: 202541076262 - Intelligent Agent for Personalized & Inclusive Learning Support System for Adaptive Learning with Agentic Strategies.

Achievements

Hackathon Runner-up: 2025 Thoothukudi District Police Cyber Hackathon (Top 2 of 800+ teams). Developed a crime prediction system forecasting 7-day probabilities with 76.72% accuracy using ARIMA and Random Forest.

Education

St. Joseph's Institute of Technology Expected 2027
B.Tech in Artificial Intelligence and Data Science CGPA: 8.4/10